VICHAKARN
MED PSU52

311-241 RESPIRATORY SYSTEM

1. ข้อสอบมีทั้งหมด 100 ข้อ
2. ข้อสอบเป็นอังกฤษทั้งหมด
3. มีเวลาในการสอบ 2.5 ชั่วโมง

หมายเหตุ: ข้อสอบฉบับนี้เกิดจากความร่วมมือของคนในรุ่นและวิชาการรุ่น MED PSU 52

1. A patient was stabbed at the level of the manubrium of sternum. Which of the following structures is most likely affected?

A. Tracheal bifurcated
B. Thoracic aorta
C. Pulmonary artery
D. Pulmonary vein
E. Azygos vein

2. A man was stabbed in the neck, 2 cm above the clavicle, and developed pneumothorax. Which of the following structures is most likely injured?

A. Cervical pleura
B. Clavicular pleura
C. Mediastinal pleura
D. Anterior pleura reflection
E. Posterior pleura reflection

3. A patient presents with herpes zoster (shingles) lesions at the umbilical region.
Which of the following dermatomes corresponds to this region?

A. T4
B. T6
C. T8
D. T10
E. T12

4. 45-year-old man has history of hoarseness for 1 month. A chest X-ray shows a mass at aortopulmonary window. Which of following left-sided nerve most likely to be compressed?

A. Vagus nerve

B. Phrenic nerve

C. 3rd intercostal nerve

D. Greater splanchnic nerve

E. Recurrent laryngeal nerve

5. A 60-year-old man has worked in a shipyard handling asbestos insulation for 35 years. He presents with fever (38.5°C), productive cough with yellow sputum, and shortness of breath for 3 days. On examination, there are crackles over the right lower lung field. Chest X-ray shows right lower-lobe consolidation. Which of the following is the most likely diagnosis?

A. Community-acquired pneumonia

B. Asbestosis

C. Silicosis

D. Lung cancer

E. Pulmonary embolism

6. A 23-year-old man presents with a gunshot wound to the right chest. Chest X-ray shows pleural effusion and partial collapse of the right lung. Which of the following is the most likely site for intrapleural fluid accumulation?

A. Posterior pleural reflection

B. Pulmonary ligament

C. Costomediastinal recess

D. Anterior pleural reflection

E. Costodiaphragmatic recess

7. A hematoxylin & eosin (H&E)-stained section of lung tissue shows the following features: Simple ciliated columnar epithelium, Abundant smooth muscle and elastic fibers in the lamina propria, No cartilage or submucosal glands Which of the following structures is most likely represented in this section?

A. Trachea

B. Bronchus

C. Bronchiole

D. Terminal bronchiole

E. Respiratory bronchiole

8. A patient is found to have inability of vocal cord abduction. Weakness of which of the following muscles is most likely responsible?

A. Cricothyroid

B. Oblique muscle

C. Thyroarytenoid

D. Posterior cricoarytenoid

E. Transverse muscle

9. 50-years-old man working in a coal mine for 30 years has been persistent cough and increase dyspnea. If a bronchial biopsy brush was done on this man. Which bronchial epithelium would be found?

A. Simple cuboidal epithelium

B. Stratified squamous epithelium

C. Pseudostratified ciliated columnar epithelium with goblet cells

D. Simple squamous epithelium

E. Transitional epithelium

10. Which of the following muscles, when weakened or paralyzed, leads to impaired vocal cord adduction?

A. Cricothyroid

B. Thyroarytenoid

C. Posterior cricoarytenoid

D. Oblique arytenoid

E. Transverse arytenoid

11. infant presented to the emergency room after respiratory distress. On exam, he was observed to have tachypnea, increased work of breathing, and decreased breath sounds. On chest X-ray, he was found to have a congenital diaphragmatic hernia. Which of these structures fail to develop?

A. Pericardioperitoneal canal

B. Pleuroperitoneal membrane

C. Septum transversum

D. Dorsal mesentery

E. Lateral body wall folds

12. During the first minute of aerobic exercise, ventilation increases rapidly before significant changes in blood gases occur. Which of the following mechanisms is primarily responsible?

A. Decreased plasma K^+

B. Activation of thermoreceptors

C. Increase in arterial PCO_2

D. Activation of proprioceptors

E. Activation of peripheral chemoreceptors

13. In the patient with pulmonary fibrosis experiencing dyspnea, which muscle is actively involved when force expiration?

A. Diaphragm

B. Scalene muscle

C. Sternocleidomastoid muscle

D. Abdominal muscle

E. External intercostal muscle

14. Which of the following lung volumes or airway parameters is typically higher in an elderly man compared with a young adult man?

A. Forced expiratory volume in 1 second (FEV_1)

B. Vital capacity (VC)

C. Forced vital capacity (FVC)

D. Airway diameter

E. Residual volume (RV)

15. During forceful expiration, such as when coughing or exercising, which of the following pressures primarily increases to push air out of the lungs?

A. Intrapleural pressure

B. Transpulmonary pressure

C. Alveolar pressure

D. Atmospheric pressure

E. None of the above

16. -

17. A 20-year-old man suffers a rib fracture that penetrates his right pleura. Which of the following is the most likely pathophysiological effect?

A. Lung collapse

B. Chest wall collapse

C. Right ventricular (RV) output decrease

D. Tidal volume decrease

E. Decreased intrapleural pressure

18. Which of the following conditions is most likely to decrease lung compliance?

A. Decreased surface area of alveoli

B. Increased alveolar surface tension

C. Decreased lung elastic recoil

D. Increased chest wall elastic recoil

E. Increased transpulmonary pressure

19. A person suffers a severe brainstem injury. On examination, they exhibit deep and slow breathing (Bradypnea with large tidal volumes). Which of the following brain areas is most likely affected?

A. Medulla oblongata

B. Upper pons

C. Thalamus

D. Midbrain

E. Cerebellum

20. When V/Q ratio of a lung region decreases (i.e., perfusion > ventilation), which of the following changes in alveolar gas tension would you expect?

A. Decreased PN_2

B. Increased PCO_2

C. Increased both PN_2 and PCO_2

D. Increased both PO_2 and PCO_2

E. Increased PO_2

21. A healthy adult experiences an increase in arterial CO_2 (hypercapnia). Which of the following responses is most likely?

A. Stop breathing

B. Moderate exercise

C. Bronchoconstriction

D. Bronchodilation

E. Decreased tidal volume

22. Which of the following mechanisms is most likely to decrease arterial PO_2 in a patient with asthma?

A. Increased alveolar PCO_2

B. Increased pulmonary blood flow

C. Decreased alveolar surface area

D. Decreased alveolar-capillary PO_2 difference

E. Increased respiratory membrane thickness

23. A preterm infant (28 weeks) presents with tachypnea, grunting, intercostal retractions, and cyanosis. Chest X-ray shows diffuse ground-glass appearance with air bronchograms. Which of the following is the most likely physiological response?

A. Normal tidal volumes with minimal effort

B. Increased work of breathing to maintain minute ventilation

C. Reduced respiratory rate to conserve energy

D. Hypercapnia without hypoxemia

E. Increased functional residual capacity (FRC)

24. In an upright, healthy adult, which of the following parameters is higher at the apex of the lung compared to the base?

A. Blood flow

B. Lung compliance

C. Transpulmonary pressure

D. Intrapleural pressure

E. Airflow

25. A 30-year-old man presents with throat pain after swallowing fish. He reports that food passage is painful. Where is the most likely site where the fish bone is lodged?

A. Pyriform recess

B. Soft palate

C. Epiglottis

D. Vocalis muscle

E. Uvula

26. An X-ray of a patient reveals a tumor located at the floor of the orbit. Which paranasal sinus is the most likely origin of this tumor?

A. Frontal sinus

B. Maxillary sinus

C. Sphenoidal sinus

D. Anterior ethmoidal air cell

E. Posterior ethmoidal air cell

27. Which of the following causes a decrease in the ventilation/perfusion ratio?

A. Atelectasis

B. Emphysema

C. Hemorrhage

D. Circulatory shock

E. Pulmonary embolism

28. Which form accounts for the majority of CO_2 transport in venous blood?

A. Dissolved CO_2 in erythrocytes

B. Carboxyhemoglobin in erythrocytes

C. H_2CO_3 in plasma

D. Dissolved CO_2 in plasma

E. HCO_3^- in plasma

29. A newborn presents with excessive drooling, choking, and cyanosis during attempts at feeding. The neonatologist attempts to pass a nasogastric tube, but it cannot be advanced into the stomach. Imaging shows coiling of the tube in the upper esophagus. What is the most likely underlying anatomical defect?

A. Pyloric stenosis

B. Esophageal web

C. Tracheoesophageal fistula

D. Tracheoesophageal septum malformation

E. Laryngeal atresia

30. A preterm neonate presents with tachypnea, grunting, and chest retractions shortly after birth. Chest X-ray shows a “ground glass” appearance. The underlying abnormality is most likely due to dysfunction or deficiency of which cells?

A. Type I pneumocytes
B. Type II pneumocytes
C. Clara cells
D. Goblet cells
E. Alveolar macrophages

31. In neonatal respiratory distress syndrome (RDS), which of the following best describes the consequence of surfactant deficiency in the lungs?

A. An increase of lung compliance
B. Stabilized alveolar volume
C. A decrease of surface tension
D. An increase of transpulmonary pressure
E. An increase of airway resistance

32. A patient with diabetes mellitus presents with altered ventilation(ketoacidosis). Which of the following receptors is primarily responsible for detecting hypoxemia and mediating increased breathing rate in this patient?

A. Juxtacapillary (J) receptors
B. Proprioceptors
C. Irritant receptors
D. Central chemoreceptors
E. Peripheral chemoreceptors

33. A 60-year-old man presents with acute pulmonary edema. He complains of dyspnea and rapid, shallow breathing. Which of the following receptors is primarily responsible for stimulating this abnormal breathing pattern?

A. Peripheral chemoreceptors

B. Juxtacapillary (J) receptors

C. Irritant receptors

D. Central chemoreceptors

E. Proprioceptors

34. A 30-year-old man develops pulmonary edema after severe hypoxemia. His chest X-ray shows bilateral infiltrates, and pulmonary capillary wedge pressure is normal. Which of the following best describes the mechanism of his pulmonary edema?

A. Increased hydrostatic pressure

B. High permeability of alveolar-capillary membrane

C. Lymphatic obstruction

D. Reduced plasma oncotic pressure

E. Airway obstruction

35. In the oxygen-hemoglobin dissociation curve, which of the following conditions increases oxygen-hemoglobin affinity (shifts the curve to the left)?

A. Hypercapnia

B. Lactic acidosis

C. Respiratory alkalosis

D. Increased 2,3-BPG

E. Increased body temperature

36. A 32-year-old man presents with chronic cough, weight loss, and night sweats. Chest X-ray shows upper lobe infiltrates with cavitation. Biopsy of the lung lesion shows granulomatous inflammation. Which of the following cells are typically seen in this condition?

A. Neutrophils and eosinophils

B. Epitheloid cells and Langhans giant cells

C. Mast cells and basophils

D. Plasma cells and B lymphocytes

E. Fibroblasts and chondrocytes

37. A patient is stabbed in the right lower chest. He develops pneumothorax. On chest X-ray, which lung region should be carefully evaluated for collapse or injury?

A. Right upper lobe

B. Right middle lobe

C. Right lower lobe

D. Left upper lobe

E. Left lower lobe

38. A 40-year-old man suffers injuries in a car accident. On examination, he is breathing shallowly and slowly. Which of the following respiratory centers is most likely affected?

A. Apneustic center (APN)

B. Pneumotaxic center (PNC)

C. Ventral respiratory group (VRG)

D. Dorsal respiratory group (DRG)

E. Reticular activating system (RAS)

39. Which of the following pathologies increases the elastic work of breathing?

A. Asthma

B. COPD

C. Emphysema

D. Pulmonary fibrosis

E. Pneumothorax

40. In which of the following situations would a healthy person most likely develop an increase in arterial PCO_2?

A. Moderate exercise

B. Mountain climbing

C. Scuba diving

D. Hyperventilation due to anxiety

E. Staying in a high-altitude area for several weeks

41. A 60-year-old man presents with progressive dyspnea and pleuritic chest pain. Physical examination shows decreased breath sounds and stony dullness on percussion at the right lower lung zone. Pleural effusion is suspected. Which chest X-ray (CXR) view would best demonstrate the presence of pleural effusion?

A. Posteroanterior (PA) view

B. Anteroposterior (AP) supine view

C. Lateral decubitus view

D. Lateral view of chest

E. Lordotic view

42. -

43. What called the volume of air remaining in the lungs at the end of normal expiration?

A. Inspiratory capacity (IC)

B. Total lung capacity (TLC)

C. Vital capacity (VC)

D. Expiratory reserve volume (ERV)

E. Functional residual capacity (FRC)

44. During the respiratory cycle, at which lung volume is airway resistance the lowest?

A. At residual volume (RV)

B. At tidal volume at rest (functional residual capacity, FRC)

C. During vagus nerve stimulation

D. During forced expiration

E. At total lung capacity (TLC)

45. -

46. A 30-year-old man has a respiratory rate (RR) of 10 breaths/min. His tidal volume (Vt) is such that total ventilation = 8 L/min. The physiological dead space (Vd) is 200 mL. Calculate the alveolar ventilation (Va)

A. 4.0 L/min
B. 5.0 L/min
C. 6.0 L/min
D. 6.4 L/min
E. 7.0 L/min

47. A 60-year-old smoker presents with progressive shortness of breath. Pulmonary function test shows decreased FEV_1 and increased lung compliance. What is the most likely diagnosis?

A. Asthma
B. Pneumonia
C. Emphysema
D. Chronic bronchitis
E. Pulmonary fibrosis

48. A 16-year-old girl with severe allergic asthma continues to have frequent exacerbations despite high-dose inhaled corticosteroids and long-acting $\beta 2$ agonists. Which of the following drugs binds to IgE and prevents it from activating mast cells and basophils?

A. Omalizumab
B. Montelukast
C. Prednisone
D. Albuterol
E. Ipratropium

49. -

50. -

51. MOA of 5-LO inhibitor

A. Neutrophil activity

B. Basophil activity

C. Eosinophil activity

D. Mast cell degranulation

E. -

52. -

53. A 45-year-old man with chronic bronchitis presents with productive cough and difficulty expectorating sputum. The physician prescribes a drug that not only helps liquefy and clear mucus, but also has the ability to stimulate pulmonary surfactant production, improving lung function. Which of the following drugs was most likely prescribed?

A. Carbocysteine

B. Ambroxol

C. Codeine

D. Dextromethorphan

E. Acetylcysteine

54. A patient has sticky mucus and a productive cough. What medication should be used?

A. Carbocysteine

B. -

C. -

D. -

E. -

55. A 30-year-old man with allergic rhinitis uses a topical nasal decongestant spray for several days. Which of the following adverse effects should be monitored or warned about?

A. Rebound congestion (rhinitis medicamentosa)

B. Sedation

C. Gastrointestinal bleeding

D. Hypoglycemia

E. Pulmonary fibrosis

56. Which effect of first-generation H1 antihistamines is responsible for reducing rhinorrhea in common cold?

A. Antimuscarinic

B. Antiserotonergic

C. Antidopaminergic

D. Alpha-adrenergic blockade

E. Beta-adrenergic stimulation

57. Regarding antihistamines, if the patient is concerned about sedation, which medication causes the least drowsiness?

A. Lorathadine

B. -

C. -

D. -

E. -

58. A 50-year-old man with COPD has an acute exacerbation with wheezing and dyspnea. After endotracheal intubation, which chest X-ray position is most appropriate to confirm the tube placement?

A. PA

B. Portable CXR, AP

C. Portable CXR, lordotic

D. Lateral

E. Left lateral decubitus

59. A 58-year-old man presents with chronic cough and upper chest pain. The physician suspects a Pancoast tumor at the apex of the right lung. Which type of chest X-ray projection is most appropriate to visualize the lung apex clearly?

A. PA (Posteroanterior)

B. AP (Anteroposterior)

C. Lateral

D. Lordotic

E. Left lateral decubitus

60. A 62-year-old male smoker undergoes a chest X-ray. There is a well-defined mass in right upper lobe (indicated by the arrow). What is most likely radiographic density of this mass?



A. Air

B. Soft tissue

C. Metallic

D. Bone

E. Fat

61. CXR shows obscured right heart border. Where is the pathology located?

A. Thoracic spine

B. Right lower lobe

C. Right middle lobe

D. Middle mediastinum

E. Posterior mediastinum

62. -

63. A 55-year-old woman with a 30-year history of smoking presents with chronic cough. A bronchial biopsy reveals squamous epithelium with keratin pearls and prominent intercellular bridges. Which of the following is the most likely radiographic pattern of her lung lesion?

A. Large hilar mass

B. Central lung nodule

C. Peripheral nodule

D. Multiple peripheral nodules

E. Central mass and a peripheral nodule

64. -

65. A 65-year-old male with a 40-pack-year smoking history presents with a central lung mass. Biopsy shows small, round blue cells with scant cytoplasm. Immunohistochemistry reveals positivity for a neuroendocrine marker and TTF-1. Which combination of IHC markers is most useful to confirm the diagnosis of small cell lung carcinoma?

A. CK7 + TTF-1

B. CK7 + CK20

C. CK20 + TTF-1

D. Chromogranin + TTF-1

E. Chromogranin + CK20

66. -

67. -

68. A 36-week gestation neonate is born with respiratory distress. Imaging shows the spleen, liver, and intestines herniated into the left thoracic cavity. What is the most likely complication affecting the lungs?

A. Congenital heart disease
B. Pulmonary agenesis
C. Pulmonary sequestration
D. Lung hypoplasia
E. Intrauterine infection

69. A 2-year-old boy presents with fever, cough, and signs of otitis media. Gram stain of a nasopharyngeal swab shows small Gram-negative coccobacilli. Culture is negative on blood agar but grows on chocolate agar. Which organism is the most likely cause?

A. Moraxella catarrhalis
B. Streptococcus pneumoniae
C. Haemophilus influenzae
D. Legionella pneumophila
E. Bordetella pertussis

70. A 25-year-old male suffers a traumatic brain injury with a complete transection above the pons. Which of the following is expected?

A. The voluntary control of breathing disappears
B. Only inhalation occurs
C. The Hering-Breuer reflex is abolished
D. Central chemoreceptors cannot control ventilation
E. Peripheral chemoreceptors cannot control ventilation

71. -

72. A 20-year-old man presents with a history of repeated severe pneumonia since childhood. Chest CT shows marked dilation of the right bronchi. Which of the following best explains the underlying cause of his condition?

A. Severe alveolar wall destruction

B. Goblet cell hyperplasia with excessive mucus secretion

C. Submucosal gland hyperplasia

D. Recurrent inflammation with bronchial wall destruction

E. Malignant tumor obstructing the right main bronchus

73. -

74. A 68-year-old presents with fever, chills, and productive cough. Sputum Gram stain shows Gram-positive, lancet-shaped diplococci. Which of the following additional laboratory findings would support the diagnosis?

A. Grows on McConkey agar, oxidase positive

B. Non-hemolytic colonies, optochin resistant

C. Grows on chocolate agar with required X and V factor

D. β-hemolytic colonies on blood agar, bacitracin sensitive

E. α-hemolytic colonies on blood agar, optochin sensitive

75. A 70-year-old man presents with high fever, productive cough, and pleuritic chest pain. Gram stain of sputum shows Gram-positive lancet-shaped diplococci. Colonies are alpha-hemolytic and optochin-sensitive. Most likely causative organism?

A. Moraxella catarrhalis

B. Streptococcus pneumoniae

C. Haemophilus influenzae

D. Klebsiella pneumoniae

E. Legionella pneumophila

76. -

77. A 12-year-old boy presents with high fever, headache, myalgia, sore throat, and cough during flu season. The physician wants to confirm influenza A infection. Which specimen is most appropriate?

A. Nasopharyngeal swab in viral transport medium

B. Throat swab

C. Sputum in sterile container

D. EDTA blood

E. Stool sample

78. A 28-year-old man presents with high fever, cough, and progressive dyspnea. He reports that during the past 2 weeks, several chickens he raised at home have died suddenly. Avian influenza infection is suspected. Which of the following is the most appropriate specimen for laboratory confirmation of the diagnosis?

A. EDTA blood

B. Sputum (lower respiratory tract specimen)

C. Serum (acute phase)

D. Throat swab

E. Urine

79. A chest X-ray shows a well-defined mass in the right middle lobe of the lung. Which of the following best describes the radiographic density of this lesion?

A. Metal

B. Soft tissue

C. Fat

D. Bone

E. Air

80. Which of the following cells is the major source of interleukin-5 (IL-5), a cytokine that plays an important role in eosinophil activation and proliferation?

A. Macrophage

B. Neutrophil

C. Eosinophil

D. Th2 lymphocyte

E. Dendritic cell

81. A patient undergoes venous blood sampling, and air accidentally enters the syringe. Which of the following changes is expected in the measured arterial blood gas?

A. Decreased PaO_2

B. Decreased $PaCO_2$

C. Increased alveolar-arterial (A–a) gradient

D. Decreased pH

E. No change

82. A patient presents with sudden dyspnea, hemoptysis, and hypoxemia after blunt chest trauma. Chest X-ray shows diffuse alveolar infiltrates. Which of the following best describes the site of bleeding?

A. Intrapleural bleeding

B. Intrapulmonary bleeding

C. Interstitial space

D. Intralobar vessels

E. Bronchial lumen

83. A 40-year-old man presents with fever, productive cough, dyspnea, and inspiratory crepitations on chest auscultation. Chest X-ray reveals consolidation in the right lower lobe. Which of the following is the main mechanism leading to hypoxemia in this patient?

A. Increased diffusion capacity of the lung

B. Increased alveolar ventilation

C. Increased pulmonary capillary perfusion without ventilation

D. Decreased V/Q ratio (V/Q mismatch due to shunt-like effect)

E. Decreased oxygen consumption in the tissues

84. A 19-year-old man has recurrent spontaneous pneumothorax. Which of the following is the most likely underlying cause?

A. Rib fracture

B. Rupture of subpleural bleb

C. Pulmonary contusion

D. Diaphragmatic injury

E. Hemothorax

85. A 30-year-old man presents with fever, weight loss, and chronic cough. Chest X-ray shows a cavitary lesion, and sputum is AFB-positive (3+). Which of the following immune mechanisms is most important for controlling this organism?

A. Complement activity

B. IgE overactivation

C. Neutrophil-mediated phagocytosis

D. NK cell-mediated cytotoxicity

E. CD4 T-cell-mediated activation of macrophages

86. A 4-year-old boy is brought to the emergency room after accidentally ingesting petroleum oil. Physical examination shows bilateral crepitations, nasal flaring, and hypoxemia (SaO_2 = 90%). After administering 100% oxygen, the hypoxemia persists. Which of the following mechanisms best explains the hypoxemia in this patient?

A. Excessive mucus production

B. Airway hypersensitivity and bronchospasm

C. Alveolar collapse and surfactant inactivation (intrapulmonary shunt)

D. Pulmonary embolism

E. Hypoventilation

87. A 35-year-old woman presents with sudden shortness of breath and anxiety. ABG: pH 7.48, $PaCO_2$ 28 mmHg, HCO_3^- 24 mEq/L. What is the primary acid-base disorder?

A. Metabolic acidosis

B. Metabolic alkalosis

C. Acute respiratory acidosis

D. Acute respiratory alkalosis

E. Mixed disorder

88. A 25-year-old man with type 1 diabetes mellitus presents to the emergency room with nausea, vomiting, abdominal pain, and fatigue for the past 2 days. He reports polyuria and polydipsia. On examination, he is tachypneic with deep, rapid respirations. Laboratory findings show: glucose 450 mg/dL,Serum ketones: positive, ABG: pH 7.33, $PaCO_2$ 28 mmHg, HCO_3^- 15 mEq/L. Which of the following best describes his primary acid-base disorder?

A. Compensated metabolic acidosis

B. Metabolic alkalosis

C. Acute respiratory acidosis

D. Acute respiratory alkalosis

E. Mixed disorder

89. A 32-year-old woman is brought to the emergency department with sudden onset of shortness of breath, chest pain, palpitations, and dizziness after receiving news of a family emergency. She is anxious, restless, and breathing rapidly at 30 breaths/min. Physical examination reveals no cyanosis and clear lung sounds. ABG shows: pH 7.51, $PaCO_2$ 28 mmHg, HCO_3^-23 mEq/L. Which of the following is the most likely acid-base disorder?

A. Acute respiratory acidosis

B. Compensated metabolic acidosis

C. Acute respiratory alkalosis

D. Metabolic alkalosis

E. Mixed disorder

90. A 65-year-old man with chronic bronchitis presents with progressive dyspnea. Arterial blood gas shows PaO_2 60 mmHg, $PaCO_2$ 40 mmHg. He is given supplemental oxygen, and his PaO_2 improves significantly. Which of the following is the most likely mechanism of his hypoxemia?

A. Hypoventilation
B. Diffusion impairment
C. V/Q mismatch
D. Right-to-left shunt
E. Low inspired O_2

91. A 3-year-old child with cyanotic congenital heart disease presents with low oxygen saturation despite oxygen therapy. Which mechanism best explains the hypoxemia?

A. Hypoventilation
B. Diffusion impairment
C. V/Q mismatch
D. Right-to-left shunt
E. Low inspired O_2

92. A 35-year-old man is brought to the emergency room after a car accident. He has multiple rib fractures involving several adjacent ribs. On physical examination, the chest wall moves inward during inspiration and outward during expiration. Which of the following best explains this paradoxical chest wall movement?

A. Flail chest due to multiple rib fractures
B. Pneumothorax causing mediastinal shift
C. Diaphragmatic rupture
D. Hemothorax compressing the lung
E. Pulmonary contusion

93. A 32-year-old woman accidentally aspirates a peanut while laughing. On chest examination, decreased breath sounds are noted on the right lower lung field. Which of the following anatomic characteristics best explains why aspirated foreign bodies are more likely to lodge in the right lower lobe bronchus?

A. Right main bronchus is longer and more horizontal than the left

B. Right main bronchus is narrower and more curved than the left

C. Right main bronchus is shorter, wider, and more vertical than the left

D. Left main bronchus is shorter and wider than the right

E. Left main bronchus is more vertical than the right

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